

Elements of Statistics and Probability

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Presentation of Data



Data

- Based on how the information is organized or presented in a dataset, there are two types of data:
 - Raw (ungrouped) data
 - Grouped data

Raw (ungrouped) Data

- In any statistical investigation, when data first collected usually appear in raw form where, information has been recorded merely in arbitrary order in which they happened to occur
- Data recorded in the sequence in which they are collected and before they are processed or ranked are called raw data
- Raw data, collected for any statistical investigation, is unable to represent the summary information
- Raw data should be represented in such a way so that we can extract the preliminary ideas about the variable(s) under study



Dealing with Raw Data

- The most frequently used methods for data representation are:
 - Tabulation
 - Frequency distribution
 - Relative frequency distribution
 - Percentage distribution
 - Graphical presentation
 - Bar chart
 - Pie chart
 - Histogram, dot plot, Box plot, stem-and-leaf plot

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Tabulation and its Principles

- A statistical method of data condensation by which we can represent summary information of one or more variables, is defined as **tabulation**
- The table should be self-explanatory. The title describing the contents of the table should be clear, concise and to the point
- The table should be as simple as possible. Two or three tables are often preferable to a large table containing too many details and variables
- The specified units of measurements for the data should be given
- Necessary code or symbols used in table should be explained in a footnote and sources of data should be mentioned

Frequency Distribution

- The number of times a particular value of a certain variable occurs in a set of observations is called the **frequency** of that value
- The manner in which the frequencies are distributed in the different classes is known as the **frequency distribution** of the values of that variable

Creating a Frequency Table

- Suppose we have the following data for which type of car BracU students drive:

Ford, Chevy, Honda, Toyota, Toyota, Nissan, Kia, Nissan, Chevy, Toyota, Honda, Chevy, Toyota, Nissan, Ford, Toyota, Nissan, Mercedes, Chevy, Ford, Nissan, Toyota, Nissan, Ford, Chevy, Toyota, Nissan, Honda, Porsche, Hyundai, Chevy, Chevy, Honda, Toyota, Chevy, Ford, Nissan, Toyota, Chevy, Honda, Chevy, Saturn, Toyota, Chevy, Chevy, Nissan, Honda, Toyota, Toyota, Nissan

Frequency Table

Table 1: Frequency table for type of car data

Category	Frequency
Ford	5
Chevy	12
Honda	6
Toyota	12
Nissan	10
Other	5
Total	50

Relative Frequency

- The **relative frequency** of a category is obtained by dividing the frequency of that category by the sum of all frequencies
- Relative frequency of a category = $\frac{\text{Frequency of that category}}{\text{Sum of all frequencies}}$
- The relative frequency shows what fractional part or proportion of the total frequency belongs to the corresponding category

Relative Frequency Table

Table 2: Relative frequency table for type of car data

Category	Frequency	Relative Frequency
Ford	5	0.10
Chevy	12	0.24
Honda	6	0.12
Toyota	12	0.24
Nissan	10	0.20
Other	5	0.10
Total	50	1.0

Percentage Distribution

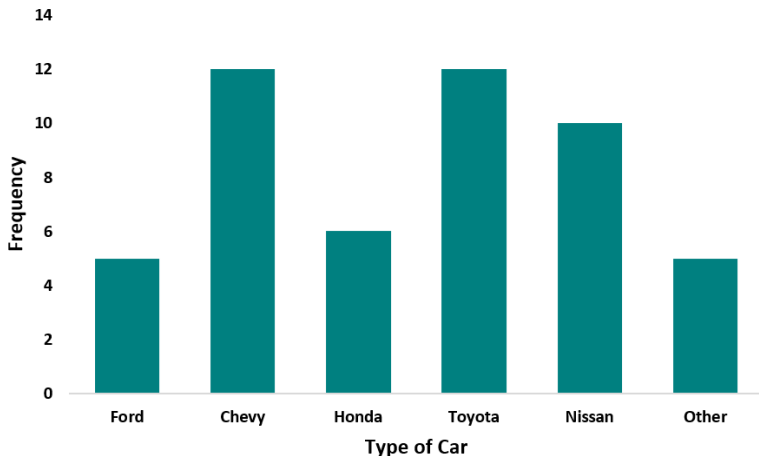
Table 3: Percentage distribution for type of car data

Category	Frequency	Relative Frequency	Percentage
Ford	5	0.10	$0.10 \times 100 = 10.0$
Chevy	12	0.24	24.0
Honda	6	0.12	12.0
Toyota	12	0.24	24.0
Nissan	10	0.20	20.0
Other	5	0.10	10.0
Total	50	1.0	100.0

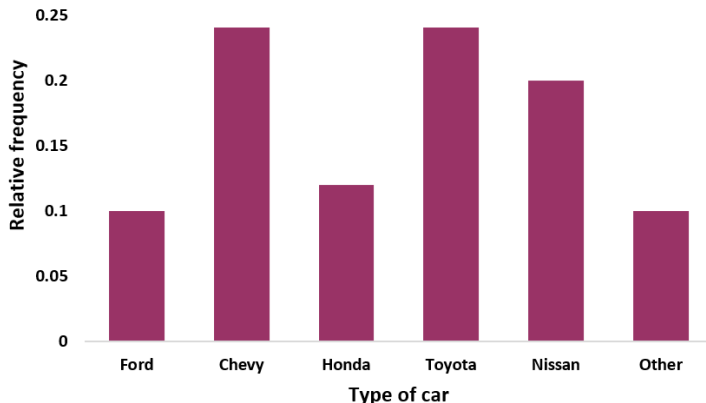
Bar Chart

- A **bar chart** is graph in which the categories are marked on X-axis the frequencies on the Y-axis (usually). The class frequencies (or relative frequency/percentage) are represented by the heights of the bars, usually there are gaps between two bars
- A bar chart is usually preferable for representing nominal or ordinal, i.e., to represent qualitative data

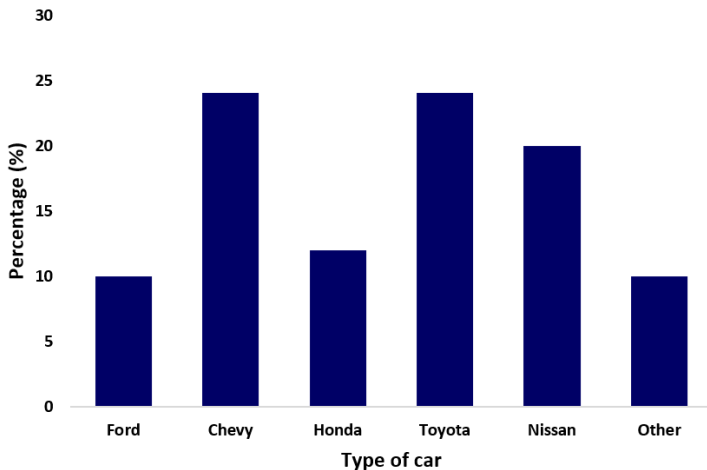
Bar Chart of Type of Car Data



Bar Chart of Type of Car Data



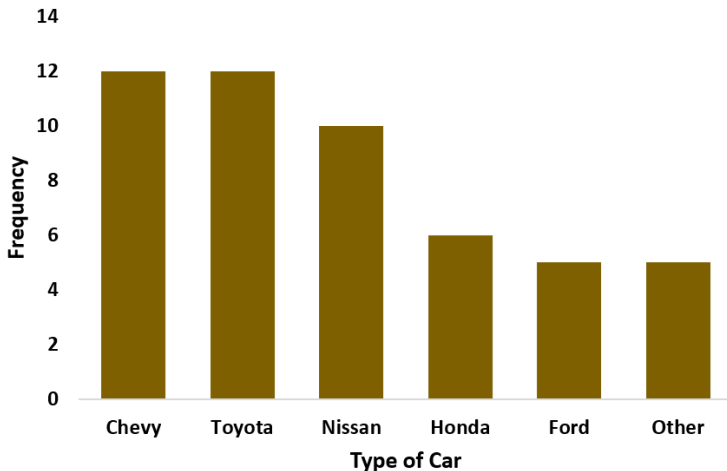
Bar Chart of Type of Car Data



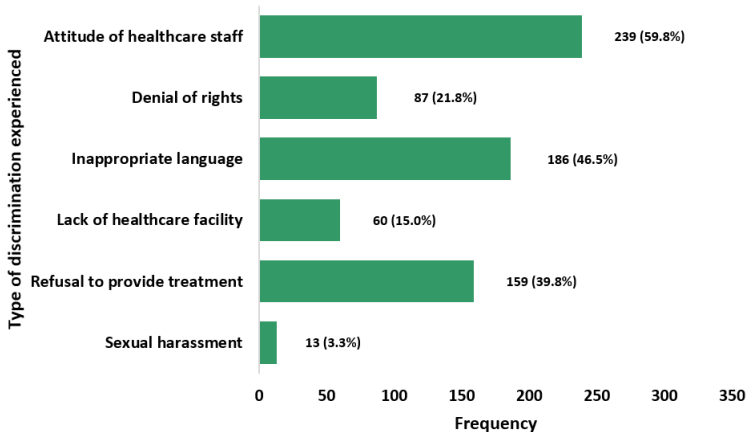
Pareto Chart

- A **pareto chart** a bar chart with the bars sorted with the highest frequencies on the left
- The advantage of Pareto charts is that we can visually see the more popular answer to the least popular
- This is especially useful in business applications, where we want to know what services your customers like the most, what processes result in more injuries, which issues employees find more important, and other type of questions like these

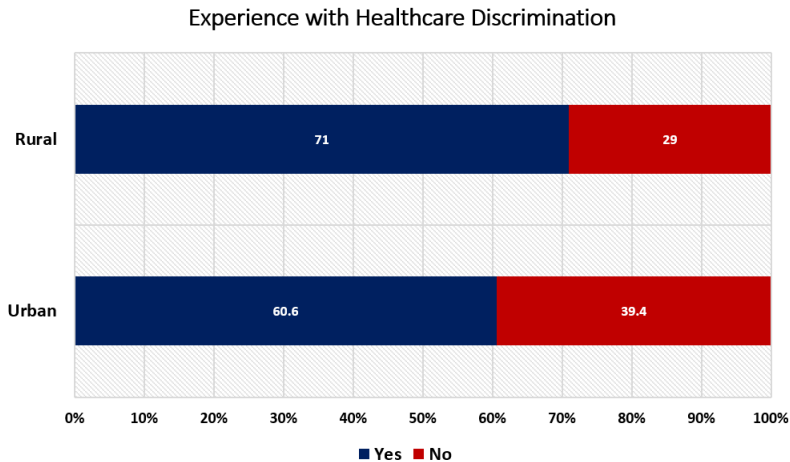
Pareto Chart for Type of Car Data



Different Types of Bar Chart

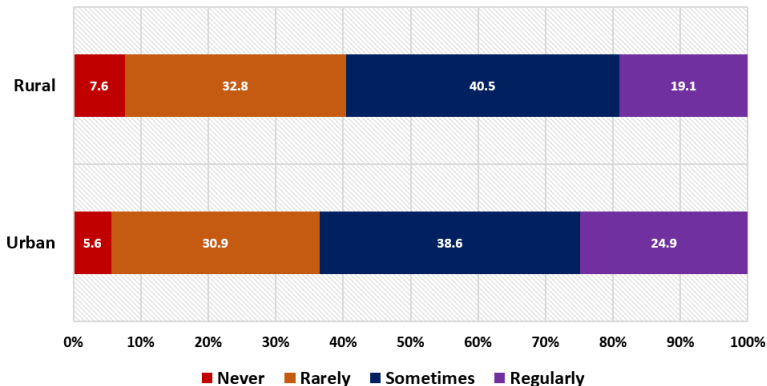


Different Types of Bar Chart (Stacked Bar Chart)

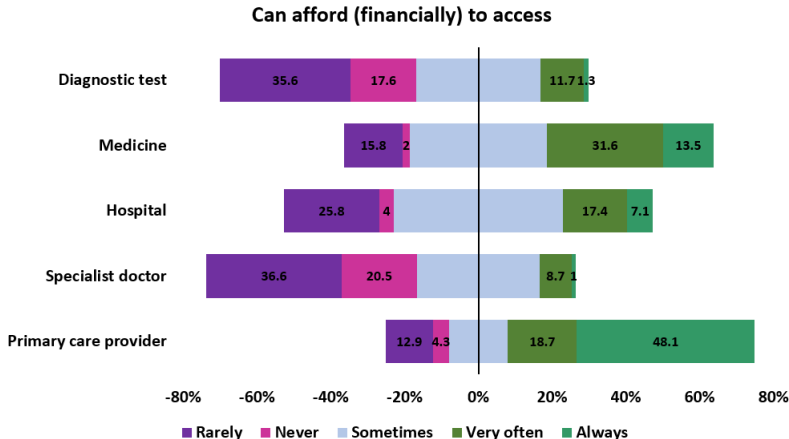


Different Types of Bar Chart (Stacked Bar Chart)

Experience with Financial Difficulties Due to Spending on Healthcare



Different Types of Bar Chart (Stacked Bar Chart)



Pie Chart

- A **pie chart** is a circle is divided proportionally to the relative frequency and portions of the circle are allocated for the different groups
- It is an appropriate representation of the data when the variable of interest is categorical (qualitative or discrete quantitative) and the total over categories is a fixed value
- If there are too many categories then a pie chart is not a good option

Constructing a Pie Chart

- Obtain a relative frequency distribution of the data
- Then multiply each relative frequency by 360° to obtain the angle measure for each category
- Now draw the pie chart using a compass, protractor, and straight edge
- The circle will be divided into wedge-shaped pieces proportional to the relative frequencies
- Label the slices with the distinct values and their relative frequencies

Pie Chart for Type of Car Data

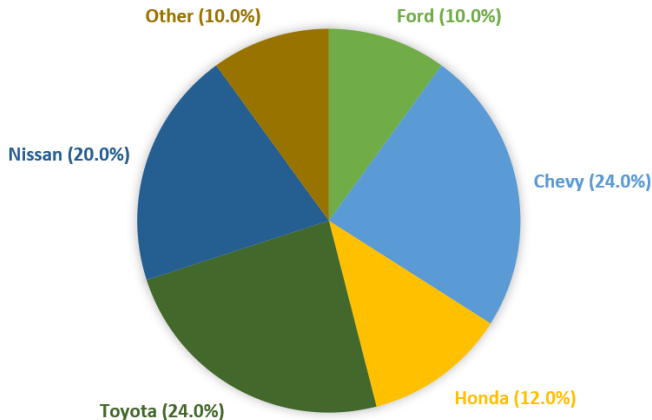
- Suppose we have the following data for which type of car students at a college drive:

Ford, Chevy, Honda, Toyota, Toyota, Nissan, Kia, Nissan, Chevy,
Toyota, Honda, Chevy, Toyota, Nissan, Ford, Toyota, Nissan,
Mercedes, Chevy, Ford, Nissan, Toyota, Nissan, Ford, Chevy,
Toyota, Nissan, Honda, Porsche, Hyundai, Chevy, Chevy, Honda,
Toyota, Chevy, Ford, Nissan, Toyota, Chevy, Honda, Chevy,
Saturn, Toyota, Chevy, Chevy, Nissan, Honda, Toyota, Toyota,
Nissan

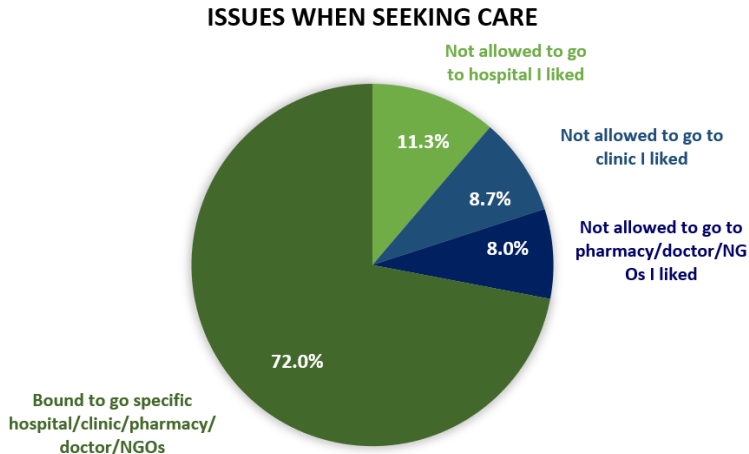
Pie Chart for Type of Car Data

Category	Frequency	Relative Frequency	Angle (in degrees)
Ford	5	0.10	$0.10 \times 360 = 36.0$
Chevy	12	0.24	86.4
Honda	6	0.12	43.2
Toyota	12	0.24	86.4
Nissan	10	0.20	72.0
Other	5	0.10	36.0
Total	50	1.0	360.0

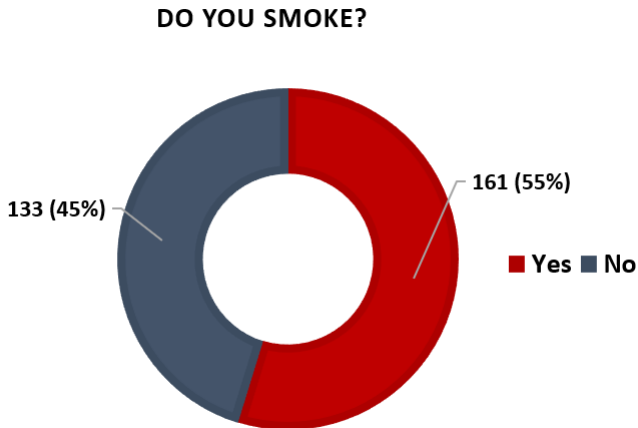
Pie Chart for Type of Car Data



Different Types of Pie Chart



Different Types of Pie Chart (Donut Chart)



Exercise 1

1. 20 BracU students were asked if they live with both parents (B), father only (F), mother only (M), or someone else (S). Data:

M B B M F S B M F M

B F B M M B B F B M

- (a) Construct a frequency distribution table
- (b) Write the relative frequencies and percentages for all categories
- (c) What percentage of the children in this sample live with their mothers only?
- (d) Draw a bar graph for the frequency distribution and a pie chart for the percentages

Exercise 2

2. A sample of 40 BracU students was selected, and these students were asked how much satisfied they were with their academic life. The responses of these employees are recorded below, where *low* represents low level of satisfaction, *moderate* means moderately satisfied and *high* stands for high level of satisfaction:

moderate	low	moderate	low	high	high	moderate	high
low	low	moderate	high	moderate	high	low	low
high	high	low	moderate	low	low	moderate	moderate
low	low	moderate	low	low	moderate	low	moderate
moderate	high	moderate	moderate	high	low	low	low

- (a) Construct a relative frequency distribution
- (b) Draw a bar chart for the frequency distribution

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Frequency Distribution

- **Frequency distribution:** A grouping of data into categories showing the number of observations in each mutually exclusive category
- Data presented in the form of a frequency distribution are called **grouped data**
- There are several method to group quantitative data into classes:
 - Single-value grouping
 - Limit grouping
 - Cut-point grouping

Single-value Grouping

- Sometimes the most appropriate way to group quantitative data is to use classes in which each class represents a single possible value
- Such classes are called single-value classes, and this method of grouping quantitative data is called **single-value** grouping
- Distinct values of the observations are used as the classes (similar to that used for qualitative data)
- Suitable for discrete data in which there are only a small number of distinct values

Example (Single-value Grouping)

- Number of bedrooms in 30 households:

2	1	3	4	4	3	2	1	1	2
1	4	4	5	1	1	2	3	4	3
1	5	3	3	5	5	1	1	2	2

Example (Single-value Grouping)

Number of bedrooms	Frequency	Relative frequency
1	9	0.3
2	6	0.2
3	6	0.2
4	6	0.2
5	3	0.1

Limit Grouping

- In this method, each class consists of a range of values
- The smallest value that could go in a class is called the lower limit of the class, and the largest value that could go in the class is called the upper limit of the class
- It is particularly useful when the data are expressed as whole numbers and there are too many distinct values to employ single-value grouping

Frequency Distribution

- **Class mark (midpoint)**: A point that divides a class into two equal parts. This is the average between the upper and lower class limits
- $$\text{Class Midpoint} = \frac{\text{Lower Limit} + \text{Upper Limit}}{2}$$
- **Class interval**: For a frequency distribution having classes of the same size, the class interval is the difference between **upper and lower limits** of a class
- $$\text{Class Interval} = \text{Upper Limit} - \text{Lower Limit}$$

Construction of Frequency Distribution

- Steps:

- (1) Determine the largest and smallest numbers in the raw data and thus find the range (difference between largest and smallest numbers)
- (2) Determine the number of classes
- (3) Calculate class interval
- (4) Determine the number of observations falling into each class interval, i.e., find the class frequencies. This is best done by using a tally or score sheet

Construction of Frequency Distribution

- The class interval used in the frequency distribution should be equal
- Sometime unequal class intervals are necessary to avoid a large number of empty or almost empty classes
- Too many classes or too few classes might not reveal the basic shape of the data
- Use your professional judgment to select the number of classes

Construction of Frequency Distribution

- **The 2^k rule:** As the number of classes, select the smallest integer, k (whole number) such that $2^k > n$
- **Sturges's rule:** Expected number of classes can be calculated using the formula: $k = 1 + 3.322 \times \log_{10} n$
- Class interval =
$$\frac{\text{highest value} - \text{lowest value}}{\text{number of classes } (k)}$$
- Consider a suitable rounded value as number of class and class interval
- It is best to not use less than 5 nor more than 15 classes in the construction of a frequency distribution

Construction of Frequency Distribution

- Prepare a table with three column headings: class interval, tally marks and frequency

Class interval	Tally	Frequency
10-20		9

- Write down the class intervals for the data set under class interval column
- Read off the data values and put a tally under tally mark column in front of the appropriate class interval
- Count the tally marks and put the figure under frequency column
- Your frequency table is prepared

Example

Table 4: Frequency Distribution of yearly Sell volume

Class interval	Tallies	Frequency
10-19		8
20-29		6
30-39		7
40-49		6
50-59		3

Some Common Terms in Limit grouping

- **Lower class limit:** The smallest value that could go on a class
- **Upper class limit:** The largest value that could go on a class
- **Class width:** The difference between the lower limit of a class and the lower limit of the next class
- **Class mark or midpoint:** The averages of the two class limits (upper and lower limits) of a class

Limitation of Limit Grouping

- The numbers in between the lower limit of a group and the upper limit of the next group does not have any specific group
- For example, for groups 30-39 and 40-49, we fail to assign a group to the value 39.5
- To overcome this issue we employ **cut-point grouping**

Cut-point Grouping

- Each class consists of a range of values
- The smallest value that could go in a class is called the lower cut-point of the class, and the smallest value that could go in the next higher class is called the upper cut-point of the class
- Note that the lower cut-point of a class is the same as its lower limit and the upper cut-point of a class is the same as the lower limit of the next higher class

Example

- Annual sell volume (in Lac TK.) of $n = 50$ companies are given following:

38.70, 42.93, 41.07, 45.66, 48.16, 15.27, 43.15, 54.88,
19.98, 33.64, 49.94, 55.45, 39.58, 47.58, 41.94, 34.39,
30.34, 36.81, 20.56, 47.33, 11.92, 14.95, 37.36, 17.67,
18.29, 12.24, 45.74, 19.39, 30.85, 23.45, 48.51, 12.45,
54.29, 57.61, 15.40, 44.10, 33.55, 32.56, 21.42, 22.12,
20.32, 45.38, 34.21, 44.84, 19.71, 15.04, 38.91, 21.59,
48.51, 28.09

- Present the data in a frequency distribution

Example

Table 5: Frequency Distribution of yearly Sell volume

Class interval	Tallies	Frequency
10- under 20		12
20- under 30		7
30- under 40		12
40- under 50		15
50- under 60		4

Some Common Terms in Limit grouping

- **Lower class cut-point:** The smallest value that could go on a class.
- **Upper class cut-point:** The smallest value that could go in the next-higher class (equivalent to the lower cut-point of the next-higher class)
- **Class width:** The difference between the cut-points of a class
- **Class mark or midpoint:** The average of the two cut-points of a class

Class Boundaries or Cut-points

- If the non-overlapping limits are considered, for example, 118-127, then 128-137, then the first class theoretically includes all measurements from 117.5 to 127.5
- These numbers, indicated briefly by the exact numbers 117.5 to 127.5 are called boundaries or **true class limits**
- Note that in case of overlapping class intervals, the class boundaries are same as the class limits

Which Grouping Method Should We Use?

- **Single-value grouping:** Use with discrete data in which there are only a small number of distinct values
- **Limit grouping:** Use when the data are expressed as whole numbers and there are too many distinct values to employ single-value grouping
- **Cut-point grouping:** Use when the data are continuous and are expressed with decimals

Histogram

- **Histogram:** A graph in which the classes are marked on the horizontal axis and the class frequencies on the vertical axis
- The frequency (relative frequency, percent) of each class is represented by a vertical bar whose height is equal to the frequency (relative frequency, percent) of that class
- The bars are drawn adjacent to each other, i.e., there is **no gap** between two bars

Value Beneath the Histogram

- For **single-value grouping**, we use the distinct values of the observations to label the bars, with each such value centered under its bar
- For **limit grouping** or **cut-point grouping**, we use the lower class limits (or, equivalently, lower class cut-points) to label the bars
- Some statisticians and technologies use class marks or class midpoints centered under the bars

Constructing a Histogram

- Obtain a frequency (relative-frequency, percent) distribution of the data
- Draw a horizontal axis on which to place the bars and a vertical axis on which to display the frequencies (relative frequencies, percents)
- For each class, construct a vertical bar whose height equals the frequency (relative frequency, percent) of that class
- Label the bars with the classes, the horizontal axis with the name of the variable, and the vertical axis with frequency (relative frequency, Percent)

Example (Histogram for Single-value Grouping)

- Number of bedrooms in 30 households:

2 1 3 4 4 3 2 1 1 2

1 4 4 5 1 1 2 3 4 3

1 5 3 3 5 5 1 1 2 2

- (a) Create a frequency distribution
- (b) Draw a histogram

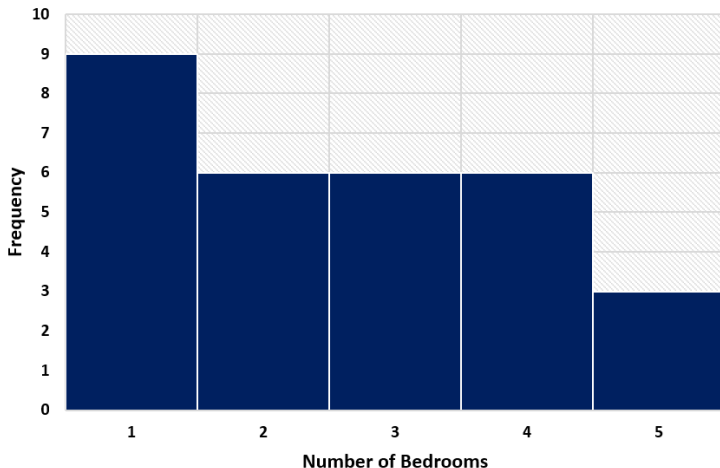
Example (Histogram for Single-value Grouping)

(a) Frequency distribution:

Number of bedrooms	Frequency	Relative frequency
1	9	0.3
2	6	0.2
3	6	0.2
4	6	0.2
5	3	0.1

Example (Histogram for Single-value Grouping)

(b) Histogram:



Example (Histogram for Limit Grouping)

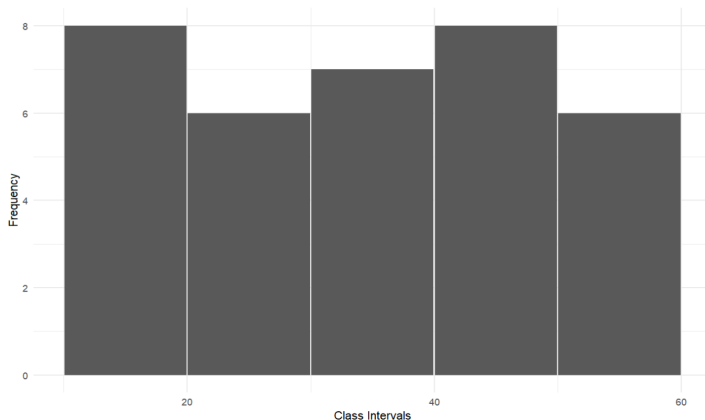


Figure 1: Histogram of yearly Sell volume

Example (Histogram for Cut-point Grouping)

- Histogram for **cut-point grouping** is same as the histogram for **limit grouping**

What to Observe in Histograms

- **Outliers**: observations that stand out from the rest for some reason
- **Center**: the “middle” of the data
- **Spread**: the range; the extent of the data; how far the values are from each other
- **Shape**: distribution pattern, skewness, symmetry, uniform etc.

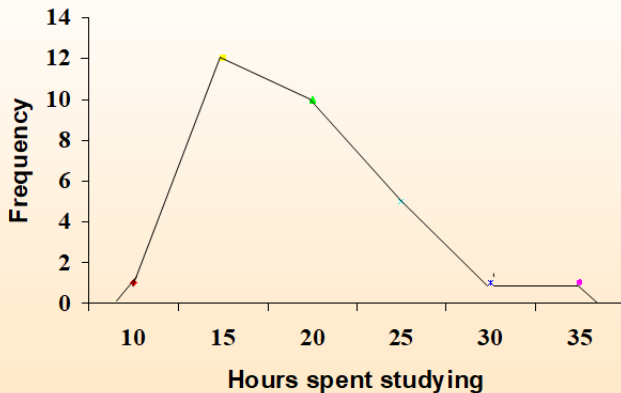
Some Insights from a Histogram

- Relative-frequency (or percent) histograms are better than frequency histograms for comparing two data sets
- The vertical scale of a frequency histogram depends on the number of observations, making comparison more difficult
- The same vertical scale is used for all relative frequency histograms (a minimum of 0 and a maximum of 1) making direct comparison easy
- Histograms based on limit or cut-point grouping can be sensitive to the choice of classes
- It is sometimes important to experiment with making different choices for the classes in order to see whether and how such choices affect the shape or other aspects of the histogram

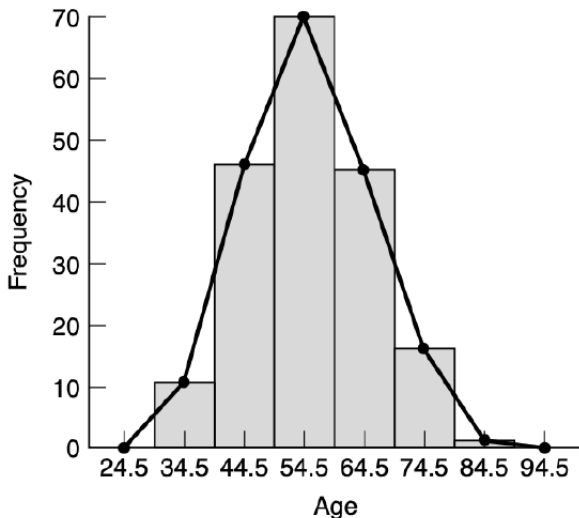
Frequency Polygon

- A **frequency polygon** consists of line segments connecting the points formed by plotting the midpoint and the class frequency for each class and then joined with X-axis at lower limit of first class and upper limit of last class

Frequency Polygon



Frequency Polygon from a Histogram



Cumulative Frequency Distribution

- A **cumulative frequency distribution** gives the total number of values that fall below the upper boundary of each class
- The **cumulative relative frequencies** are obtained by dividing the cumulative frequencies by the total number of observations in the data set
- The **cumulative percentages** are obtained by multiplying the cumulative relative frequencies by 100

Cumulative Frequency Distribution

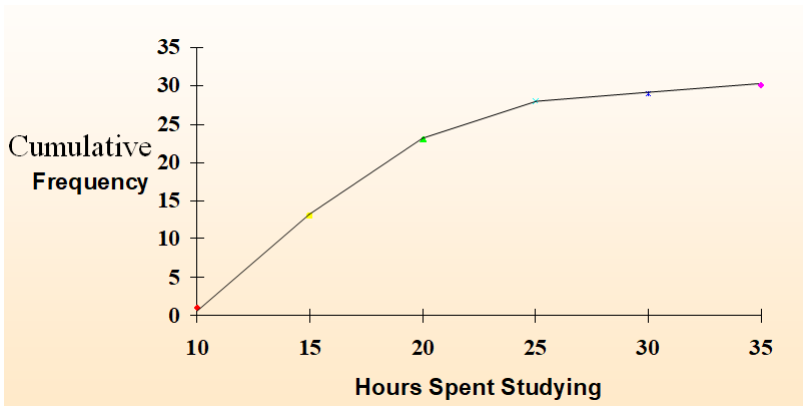
Table 6: Frequency Distribution of yearly Sell volume

Class interval	Frequency	Cumulative frequency
10-19	8	8
20-29	6	$8 + 6 = 14$
30-39	7	$8 + 6 + 7 = 21$
40-49	6	$8 + 6 + 7 + 6 = 27$
50-59	3	$8 + 6 + 7 + 6 + 3 = 30$
Total	30	

Cumulative Frequency Curve (Ogive)

- A cumulative frequency curve (ogive) is a smooth curve obtained by joining the points formed by plotting upper limit (less than type) or lower limit (more than type) of and the cumulative frequency of each class
- It is used to determine how many or what proportion of the data values are below or above a certain value
- Ogive plot is usually used to determine the values of certain quantities such as median, quartile, percentile

Cumulative Frequency Curve (Ogive)



Exercise

- The following data shows profit on 180 vehicles sold at Applewood auto group

Profit	Midpoint	Frequency
\$ 200 up to \$ 600	\$ 400	8
600 up to 1,000	800	11
1,000 up to 1,400	1,200	23
1,400 up to 1,800	1,600	38
1,800 up to 2,200	2,000	45
2,200 up to 2,600	2,400	32
2,600 up to 3,000	2,800	19
3,000 up to 3,400	3,200	4
Total		180

- Draw a frequency polygon
- Draw a cumulative frequency curve

Exercise (Solution)

(a) Frequency polygon:

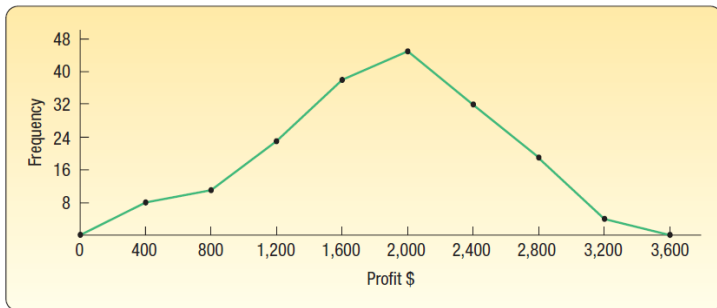


Figure 2: Frequency Polygon of Profit on 180 Vehicles Sold at Applewood Auto Group

Exercise (Solution)

(b) Cumulative frequency distribution:

Profit	Cumulative Frequency	Cumulative Relative Frequency
Less than \$ 600	8	$8/180 = 0.044 = 4.4\%$
Less than \$1,000	19	$19/180 = 0.106 = 10.6\%$
Less than \$1,400	42	$42/180 = 0.233 = 23.3\%$
Less than \$1,800	80	$80/180 = 0.444 = 44.4\%$
Less than \$2,200	125	$125/180 = 0.694 = 69.4\%$
Less than \$2,600	157	$157/180 = 0.872 = 87.2\%$
Less than \$3,000	176	$176/180 = 0.978 = 97.8\%$
Less than \$3,400	180	$180/180 = 1.000 = 100\%$

Exercise (Solution)

(b) Cumulative frequency curve:

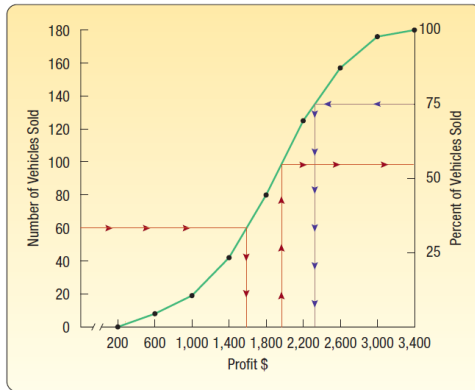


Figure 3: Ogive plot of Profit on 180 Vehicles Sold at Applewood

Auto Group

Thank You